

Geometry of Particle Methods

Definition (Fisher-Rao Gradient)

The Fisher-Rao gradient of $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$ at the point $\mu \in \mathcal{P}_2(\mathcal{M})$ is (when it exists and is square integrable) the function,

$$\nabla^{FR} \mathcal{F}[\mu](x) = \delta \mathcal{F}[\mu](x) - \int \delta \mathcal{F}[\mu](x) \mu(dx)$$

Intuition: steepest descent dynamics via mass reweighting.

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Analogous formulas hold for $M_+(\mathcal{M})$ (positive msrs. on $\mathcal{M}$)!